

	Scene 1 $\beta = 90^\circ$					Scene 1 $\beta = 45^\circ$					Scene 1 $\beta = 30^\circ$					Scene 1 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.718		5	1	1	0.718		5	1	1	0.718		5	1	1	0.718	
	5	1	2	0.718		5	1	2	0.682		5	1	2	0.682		5	1	2	0.720	
	5	1	3	0.718		5	1	3	0.682		5	1	3	0.682		5	1	3	0.682	
	5	2	1	0.718		5	2	1	0.718		5	2	1	0.718		5	2	1	0.718	
	5	2	2	0.718		5	2	2	0.713		5	2	2	0.713		5	2	2	0.713	
	5	2	3	0.719		5	2	3	0.712		5	2	3	0.712		5	2	3	0.712	
	5	3	1	0.718		5	3	1	0.718		5	3	1	0.718		5	3	1	0.718	
	5	3	2	0.718		5	3	2	0.714		5	3	2	0.714		5	3	2	0.714	
	5	3	3	0.718		5	3	3	0.709		5	3	3	0.713		5	3	3	0.711	
	15	1	1	0.716		15	1	1	0.716		15	1	1	0.722		15	1	1	0.722	
	15	1	2	0.716		15	1	2	0.680		15	1	2	0.680		15	1	2	0.680	
	15	1	3	0.716		15	1	3	0.680		15	1	3	0.680		15	1	3	0.680	
	15	2	1	0.716		15	2	1	0.716		15	2	1	0.714		15	2	1	0.714	
	15	2	2	0.718		15	2	2	0.682		15	2	2	0.682		15	2	2	0.682	
	15	2	3	0.718		15	2	3	0.682		15	2	3	0.682		15	2	3	0.682	
	15	3	1	0.717		15	3	1	0.717		15	3	1	0.722		15	3	1	0.722	
	15	3	2	0.719		15	3	2	0.683		15	3	2	0.683		15	3	2	0.683	
	15	3	3	0.717		15	3	3	0.681		15	3	3	0.681		15	3	3	0.681	
	30	1	1	0.713		30	1	1	0.713		30	1	1	0.713		30	1	1	0.718	
	30	1	2	0.713		30	1	2	0.677		30	1	2	0.677		30	1	2	0.677	
	30	1	3	0.713		30	1	3	0.677		30	1	3	0.677		30	1	3	0.677	
	30	2	1	0.713		30	2	1	0.713		30	2	1	0.713		30	2	1	0.716	
	30	2	2	0.716		30	2	2	0.716		30	2	2	0.716		30	2	2	0.722	
	30	2	3	0.717		30	2	3	0.717		30	2	3	0.717		30	2	3	0.723	
	30	3	1	0.716		30	3	1	0.716		30	3	1	0.716		30	3	1	0.717	
	30	3	2	0.717		30	3	2	0.717		30	3	2	0.717		30	3	2	0.723	
	30	3	3	0.719		30	3	3	0.721		30	3	3	0.721		30	3	3	0.725	
	45	1	1	0.704		45	1	1	0.704		45	1	1	0.704		45	1	1	0.704	
	45	1	2	0.704		45	1	2	0.704		45	1	2	0.704		45	1	2	0.669	
	45	1	3	0.704		45	1	3	0.711		45	1	3	0.711		45	1	3	0.682	
	45	2	1	0.712		45	2	1	0.712		45	2	1	0.712		45	2	1	0.712	
	45	2	2	0.716		45	2	2	0.714		45	2	2	0.716		45	2	2	0.681	
	45	2	3	0.716		45	2	3	0.716		45	2	3	0.716		45	2	3	0.722	
	45	3	1	0.712		45	3	1	0.712		45	3	1	0.712		45	3	1	0.712	
	45	3	2	0.704		45	3	2	0.704		45	3	2	0.704		45	3	2	0.704	
	45	3	3	0.719		45	3	3	0.719		45	3	3	0.719		45	3	3	0.719	

	Scene 2 $\beta = 90^\circ$					Scene 2 $\beta = 45^\circ$					Scene 2 $\beta = 30^\circ$					Scene 2 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.603		5	1	1	0.603		5	1	1	0.603		5	1	1	0.603	
	5	1	2	0.603		5	1	2	0.573		5	1	2	0.573		5	1	2	0.605	
	5	1	3	0.603		5	1	3	0.605		5	1	3	0.605		5	1	3	0.605	
	5	2	1	0.603		5	2	1	0.603		5	2	1	0.603		5	2	1	0.603	
	5	2	2	0.607		5	2	2	0.577		5	2	2	0.577		5	2	2	0.577	
	5	2	3	0.609		5	2	3	0.579		5	2	3	0.579		5	2	3	0.579	
	5	3	1	0.608		5	3	1	0.608		5	3	1	0.608		5	3	1	0.608	
	5	3	2	0.609		5	3	2	0.579		5	3	2	0.579		5	3	2	0.579	
	5	3	3	0.610		5	3	3	0.580		5	3	3	0.580		5	3	3	0.580	
	15	1	1	0.610		15	1	1	0.610		15	1	1	0.610		15	1	1	0.610	
	15	1	2	0.610		15	1	2	0.580		15	1	2	0.580		15	1	2	0.580	
	15	1	3	0.610		15	1	3	0.580		15	1	3	0.580		15	1	3	0.580	
	15	2	1	0.610		15	2	1	0.610		15	2	1	0.610		15	2	1	0.610	
	15	2	2	0.610		15	2	2	0.580		15	2	2	0.580		15	2	2	0.580	
	15	2	3	0.611		15	2	3	0.580		15	2	3	0.580		15	2	3	0.580	
	15	3	1	0.611		15	3	1	0.611		15	3	1	0.611		15	3	1	0.611	
	15	3	2	0.612		15	3	2	0.581		15	3	2	0.581		15	3	2	0.581	
	15	3	3	0.612		15	3	3	0.581		15	3	3	0.581		15	3	3	0.581	
	30	1	1	0.608		30	1	1	0.608		30	1	1	0.608		30	1	1	0.608	
	30	1	2	0.608		30	1	2	0.578		30	1	2	0.578		30	1	2	0.578	
	30	1	3	0.608		30	1	3	0.578		30	1	3	0.578		30	1	3	0.578	
	30	2	1	0.608		30	2	1	0.608		30	2	1	0.608		30	2	1	0.608	
	30	2	2	0.609		30	2	2	0.609		30	2	2	0.607		30	2	2	0.639	
	30	2	3	0.611		30	2	3	0.611		30	2	3	0.611		30	2	3	0.642	
	30	3	1	0.610		30	3	1	0.610		30	3	1	0.610		30	3	1	0.610	
	30	3	2	0.612		30	3	2	0.612		30	3	2	0.612		30	3	2	0.643	
	30	3	3	0.613		30	3	3	0.613		30	3	3	0.617		30	3	3	0.644	
	45	1	1	0.600		45	1	1	0.600		45	1	1	0.600		45	1	1	0.600	
	45	1	2	0.600		45	1	2	0.606		45	1	2	0.609		45	1	2	0.630	
	45	1	3	0.600		45	1	3	0.593		45	1	3	0.598		45	1	3	0.630	
	45	2	1	0.600		45	2	1	0.600		45	2	1	0.600		45	2	1	0.600	
	45	2	2	0.610		45	2	2	0.610		45	2	2	0.610		45	2	2	0.641	
	45	2	3	0.612		45	2	3	0.612		45	2	3	0.612		45	2	3	0.612	
	45	3	1	0.609		45	3	1	0.609		45	3	1	0.609		45	3	1	0.609	
	45	3	2	0.610		45	3	2	0.608		45	3	2	0.610		45	3	2	0.610	
	45	3	3	0.612		45	3	3	0.607		45	3	3	0.607		45	3	3	0.608	

	Scene 3 $\beta = 90^\circ$					Scene 3 $\beta = 45^\circ$					Scene 3 $\beta = 30^\circ$					Scene 3 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.604		5	1	1	0.615		5	1	1	0.619		5	1	1	0.620	
	5	1	2	0.604		5	1	2	0.634		5	1	2	0.634		5	1	2	0.582	
	5	1	3	0.604		5	1	3	0.607		5	1	3	0.616		5	1	3	0.627	
	5	2	1	0.604		5	2	1	0.623		5	2	1	0.622		5	2	1	0.618	
	5	2	2	0.623		5	2	2	0.654		5	2	2	0.654		5	2	2	0.654	
	5	2	3	0.635		5	2	3	0.667		5	2	3	0.667		5	2	3	0.667	
	5	3	1	0.622		5	3	1	0.643		5	3	1	0.643		5	3	1	0.638	
	5	3	2	0.638		5	3	2	0.670		5	3	2	0.670		5	3	2	0.670	
	5	3	3	0.644		5	3	3	0.676		5	3	3	0.676		5	3	3	0.676	
	15	1	1	0.718		15	1	1	0.725		15	1	1	0.725		15	1	1	0.725	
	15	1	2	0.718		15	1	2	0.754		15	1	2	0.754		15	1	2	0.754	
	15	1	3	0.718		15	1	3	0.754		15	1	3	0.754		15	1	3	0.754	
	15	2	1	0.747		15	2	1	0.749		15	2	1	0.749		15	2	1	0.749	
	15	2	2	0.734		15	2	2	0.771		15	2	2	0.771		15	2	2	0.771	
	15	2	3	0.718		15	2	3	0.754		15	2	3	0.754		15	2	3	0.754	
	15	3	1	0.747		15	3	1	0.747		15	3	1	0.747		15	3	1	0.747	
	15	3	2	0.748		15	3	2	0.785		15	3	2	0.785		15	3	2	0.785	
	15	3	3	0.739		15	3	3	0.776		15	3	3	0.776		15	3	3	0.776	
	30	1	1	0.699		30	1	1	0.706		30	1	1	0.708		30	1	1	0.708	
	30	1	2	0.699		30	1	2	0.734		30	1	2	0.734		30	1	2	0.734	
	30	1	3	0.699		30	1	3	0.734		30	1	3	0.734		30	1	3	0.734	
	30	2	1	0.746		30	2	1	0.746		30	2	1	0.741		30	2	1	0.737	
	30	2	2	0.736		30	2	2	0.731		30	2	2	0.752		30	2	2	0.699	
	30	2	3	0.699		30	2	3	0.707		30	2	3	0.734		30	2	3	0.664	
	30	3	1	0.747		30	3	1	0.747		30	3	1	0.745		30	3	1	0.743	
	30	3	2	0.747		30	3	2	0.747		30	3	2	0.784		30	3	2	0.710	
	30	3	3	0.745		30	3	3	0.745		30	3	3	0.746		30	3	3	0.708	
	45	1	1	0.671		45	1	1	0.705		45	1	1	0.705		45	1	1	0.705	
	45	1	2	0.671		45	1	2	0.705		45	1	2	0.705		45	1	2	0.637	
	45	1	3	0.671		45	1	3	0.705		45	1	3	0.705		45	1	3	0.637	
	45	2	1	0.744		45	2	1	0.781		45	2	1	0.781		45	2	1	0.781	
	45	2	2	0.747		45	2	2	0.784		45	2	2	0.784		45	2	2	0.710	
	45	2	3	0.671		45	2	3	0.686		45	2	3	0.702		45	2	3	0.637	
	45	3	1	0.743		45	3	1	0.780		45	3	1	0.780		45	3	1	0.780	
	45	3	2	0.746		45	3	2	0.783		45	3	2	0.783		45	3	2	0.783	
	45	3	3	0.732		45	3	3	0.769		45	3	3	0.769		45	3	3	0.769	

	Scene 4 $\beta = 90^\circ$					Scene 4 $\beta = 45^\circ$					Scene 4 $\beta = 30^\circ$					Scene 4 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.603		5	1	1	0.610		5	1	1	0.611		5	1	1	0.611	
	5	1	2	0.603		5	1	2	0.573		5	1	2	0.573		5	1	2	0.573	
	5	1	3	0.603		5	1	3	0.599		5	1	3	0.598		5	1	3	0.596	
	5	2	1	0.604		5	2	1	0.613		5	2	1	0.614		5	2	1	0.612	
	5	2	2	0.604		5	2	2	0.574		5	2	2	0.574		5	2	2	0.574	
	5	2	3	0.603		5	2	3	0.573		5	2	3	0.573		5	2	3	0.573	
	5	3	1	0.604		5	3	1	0.613		5	3	1	0.612		5	3	1	0.612	
	5	3	2	0.605		5	3	2	0.575		5	3	2	0.575		5	3	2	0.575	
	5	3	3	0.604		5	3	3	0.574		5	3	3	0.574		5	3	3	0.574	
	15	1	1	0.602		15	1	1	0.629		15	1	1	0.632		15	1	1	0.632	
	15	1	2	0.602		15	1	2	0.572		15	1	2	0.572		15	1	2	0.572	
	15	1	3	0.602		15	1	3	0.572		15	1	3	0.572		15	1	3	0.572	
	15	2	1	0.602		15	2	1	0.629		15	2	1	0.631		15	2	1	0.625	
	15	2	2	0.604		15	2	2	0.574		15	2	2	0.574		15	2	2	0.574	
	15	2	3	0.603		15	2	3	0.573		15	2	3	0.573		15	2	3	0.573	
	15	3	1	0.603		15	3	1	0.633		15	3	1	0.632		15	3	1	0.626	
	15	3	2	0.604		15	3	2	0.574		15	3	2	0.574		15	3	2	0.574	
	15	3	3	0.604		15	3	3	0.574		15	3	3	0.574		15	3	3	0.574	
	30	1	1	0.599		30	1	1	0.629		30	1	1	0.629		30	1	1	0.629	
	30	1	2	0.599		30	1	2	0.569		30	1	2	0.569		30	1	2	0.569	
	30	1	3	0.599		30	1	3	0.569		30	1	3	0.569		30	1	3	0.569	
	30	2	1	0.599		30	2	1	0.627		30	2	1	0.628		30	2	1	0.628	
	30	2	2	0.602		30	2	2	0.595		30	2	2	0.594		30	2	2	0.632	
	30	2	3	0.604		30	2	3	0.605		30	2	3	0.600		30	2	3	0.634	
	30	3	1	0.603		30	3	1	0.632		30	3	1	0.629		30	3	1	0.629	
	30	3	2	0.604		30	3	2	0.597		30	3	2	0.598		30	3	2	0.634	
	30	3	3	0.604		30	3	3	0.599		30	3	3	0.607		30	3	3	0.634	
	45	1	1	0.577		45	1	1	0.592		45	1	1	0.592		45	1	1	0.592	
	45	1	2	0.577		45	1	2	0.606		45	1	2	0.606		45	1	2	0.606	
	45	1	3	0.577		45	1	3	0.601		45	1	3	0.606		45	1	3	0.606	
	45	2	1	0.577		45	2	1	0.577		45	2	1	0.585		45	2	1	0.584	
	45	2	2	0.601		45	2	2	0.601		45	2	2	0.600		45	2	2	0.597	
	45	2	3	0.604		45	2	3	0.605		45	2	3	0.619		45	2	3	0.619	
	45	3	1	0.598		45	3	1	0.598		45	3	1	0.601		45	3	1	0.598	
	45	3	2	0.604		45	3	2	0.605		45	3	2	0.605		45	3	2	0.604	
	45	3	3	0.604		45	3	3	0.604		45	3	3	0.608		45	3	3	0.607	

	Scene 5 $\beta = 90^\circ$					Scene 5 $\beta = 45^\circ$					Scene 5 $\beta = 30^\circ$					Scene 5 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.605		5	1	1	0.605		5	1	1	0.605		5	1	1	0.605	
	5	1	2	0.605		5	1	2	0.635		5	1	2	0.635		5	1	2	0.635	
	5	1	3	0.605		5	1	3	0.635		5	1	3	0.635		5	1	3	0.635	
	5	2	1	0.605		5	2	1	0.605		5	2	1	0.604		5	2	1	0.604	
	5	2	2	0.631		5	2	2	0.663		5	2	2	0.663		5	2	2	0.663	
	5	2	3	0.620		5	2	3	0.651		5	2	3	0.651		5	2	3	0.651	
	5	3	1	0.616		5	3	1	0.616		5	3	1	0.617		5	3	1	0.617	
	5	3	2	0.637		5	3	2	0.669		5	3	2	0.669		5	3	2	0.669	
	5	3	3	0.625		5	3	3	0.656		5	3	3	0.656		5	3	3	0.656	
	15	1	1	0.686		15	1	1	0.686		15	1	1	0.687		15	1	1	0.689	
	15	1	2	0.686		15	1	2	0.720		15	1	2	0.720		15	1	2	0.720	
	15	1	3	0.686		15	1	3	0.720		15	1	3	0.720		15	1	3	0.720	
	15	2	1	0.689		15	2	1	0.689		15	2	1	0.689		15	2	1	0.688	
	15	2	2	0.686		15	2	2	0.720		15	2	2	0.720		15	2	2	0.720	
	15	2	3	0.687		15	2	3	0.721		15	2	3	0.721		15	2	3	0.721	
	15	3	1	0.691		15	3	1	0.692		15	3	1	0.692		15	3	1	0.694	
	15	3	2	0.689		15	3	2	0.723		15	3	2	0.723		15	3	2	0.723	
	15	3	3	0.689		15	3	3	0.723		15	3	3	0.723		15	3	3	0.723	
	30	1	1	0.651		30	1	1	0.651		30	1	1	0.651		30	1	1	0.652	
	30	1	2	0.651		30	1	2	0.684		30	1	2	0.684		30	1	2	0.684	
	30	1	3	0.651		30	1	3	0.684		30	1	3	0.684		30	1	3	0.684	
	30	2	1	0.690		30	2	1	0.690		30	2	1	0.690		30	2	1	0.693	
	30	2	2	0.690		30	2	2	0.690		30	2	2	0.692		30	2	2	0.656	
	30	2	3	0.651		30	2	3	0.651		30	2	3	0.651		30	2	3	0.618	
	30	3	1	0.689		30	3	1	0.689		30	3	1	0.689		30	3	1	0.689	
	30	3	2	0.690		30	3	2	0.690		30	3	2	0.690		30	3	2	0.656	
	30	3	3	0.664		30	3	3	0.664		30	3	3	0.660		30	3	3	0.631	
	45	1	1	0.682		45	1	1	0.682		45	1	1	0.692		45	1	1	0.693	
	45	1	2	0.682		45	1	2	0.700		45	1	2	0.690		45	1	2	0.648	
	45	1	3	0.682		45	1	3	0.716		45	1	3	0.706		45	1	3	0.648	
	45	2	1	0.690		45	2	1	0.681		45	2	1	0.688		45	2	1	0.686	
	45	2	2	0.690		45	2	2	0.690		45	2	2	0.690		45	2	2	0.688	
	45	2	3	0.682		45	2	3	0.682		45	2	3	0.683		45	2	3	0.682	
	45	3	1	0.689		45	3	1	0.701		45	3	1	0.694		45	3	1	0.693	
	45	3	2	0.690		45	3	2	0.692		45	3	2	0.691		45	3	2	0.689	
	45	3	3	0.689		45	3	3	0.691		45	3	3	0.690		45	3	3	0.687	

	Scene 6 $\beta = 90^\circ$					Scene 6 $\beta = 45^\circ$					Scene 6 $\beta = 30^\circ$					Scene 6 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.703		5	1	1	0.703		5	1	1	0.703		5	1	1	0.704	
	5	1	2	0.703		5	1	2	0.721		5	1	2	0.725		5	1	2	0.725	
	5	1	3	0.703		5	1	3	0.726		5	1	3	0.728		5	1	3	0.729	
	5	2	1	0.703		5	2	1	0.705		5	2	1	0.706		5	2	1	0.702	
	5	2	2	0.703		5	2	2	0.715		5	2	2	0.712		5	2	2	0.706	
	5	2	3	0.703		5	2	3	0.713		5	2	3	0.712		5	2	3	0.706	
	5	3	1	0.704		5	3	1	0.704		5	3	1	0.704		5	3	1	0.704	
	5	3	2	0.704		5	3	2	0.719		5	3	2	0.722		5	3	2	0.716	
	5	3	3	0.704		5	3	3	0.717		5	3	3	0.719		5	3	3	0.714	
	15	1	1	0.703		15	1	1	0.704		15	1	1	0.711		15	1	1	0.711	
	15	1	2	0.703		15	1	2	0.713		15	1	2	0.713		15	1	2	0.713	
	15	1	3	0.703		15	1	3	0.705		15	1	3	0.704		15	1	3	0.704	
	15	2	1	0.704		15	2	1	0.704		15	2	1	0.701		15	2	1	0.701	
	15	2	2	0.704		15	2	2	0.704		15	2	2	0.701		15	2	2	0.701	
	15	2	3	0.703		15	2	3	0.704		15	2	3	0.700		15	2	3	0.702	
	15	3	1	0.704		15	3	1	0.711		15	3	1	0.711		15	3	1	0.711	
	15	3	2	0.703		15	3	2	0.715		15	3	2	0.711		15	3	2	0.711	
	15	3	3	0.704		15	3	3	0.710		15	3	3	0.708		15	3	3	0.709	
	30	1	1	0.702		30	1	1	0.706		30	1	1	0.715		30	1	1	0.715	
	30	1	2	0.702		30	1	2	0.710		30	1	2	0.698		30	1	2	0.706	
	30	1	3	0.702		30	1	3	0.727		30	1	3	0.702		30	1	3	0.712	
	30	2	1	0.703		30	2	1	0.703		30	2	1	0.700		30	2	1	0.700	
	30	2	2	0.703		30	2	2	0.703		30	2	2	0.700		30	2	2	0.701	
	30	2	3	0.703		30	2	3	0.704		30	2	3	0.703		30	2	3	0.704	
	30	3	1	0.702		30	3	1	0.704		30	3	1	0.706		30	3	1	0.710	
	30	3	2	0.703		30	3	2	0.703		30	3	2	0.703		30	3	2	0.703	
	30	3	3	0.704		30	3	3	0.705		30	3	3	0.712		30	3	3	0.712	
	45	1	1	0.702		45	1	1	0.702		45	1	1	0.709		45	1	1	0.711	
	45	1	2	0.702		45	1	2	0.719		45	1	2	0.703		45	1	2	0.736	
	45	1	3	0.702		45	1	3	0.737		45	1	3	0.719		45	1	3	0.737	
	45	2	1	0.702		45	2	1	0.702		45	2	1	0.704		45	2	1	0.696	
	45	2	2	0.702		45	2	2	0.706		45	2	2	0.710		45	2	2	0.707	
	45	2	3	0.702		45	2	3	0.704		45	2	3	0.705		45	2	3	0.701	
	45	3	1	0.703		45	3	1	0.704		45	3	1	0.704		45	3	1	0.703	
	45	3	2	0.704		45	3	2	0.706		45	3	2	0.705		45	3	2	0.705	
	45	3	3	0.704		45	3	3	0.705		45	3	3	0.706		45	3	3	0.706	

Scene 7 $\beta = 90^\circ$				Scene 7 $\beta = 45^\circ$				Scene 7 $\beta = 30^\circ$				Scene 7 $\beta = 15^\circ$			
α	r	s	IoU	α	r	s	IoU	α	r	s	IoU	α	r	s	IoU
5	1	1	0.620	5	1	1	0.627	5	1	1	0.628	5	1	1	0.628
5	1	2	0.620	5	1	2	0.651	5	1	2	0.651	5	1	2	0.651
5	1	3	0.620	5	1	3	0.651	5	1	3	0.651	5	1	3	0.651
5	2	1	0.620	5	2	1	0.629	5	2	1	0.630	5	2	1	0.628
5	2	2	0.633	5	2	2	0.665	5	2	2	0.665	5	2	2	0.665
5	2	3	0.625	5	2	3	0.656	5	2	3	0.656	5	2	3	0.656
5	3	1	0.625	5	3	1	0.635	5	3	1	0.635	5	3	1	0.635
5	3	2	0.635	5	3	2	0.667	5	3	2	0.667	5	3	2	0.667
5	3	3	0.626	5	3	3	0.657	5	3	3	0.657	5	3	3	0.657
15	1	1	0.634	15	1	1	0.634	15	1	1	0.634	15	1	1	0.634
15	1	2	0.634	15	1	2	0.636	15	1	2	0.637	15	1	2	0.637
15	1	3	0.634	15	1	3	0.666	15	1	3	0.666	15	1	3	0.666
15	2	1	0.640	15	2	1	0.640	15	2	1	0.640	15	2	1	0.640
15	2	2	0.634	15	2	2	0.606	15	2	2	0.606	15	2	2	0.606
15	2	3	0.638	15	2	3	0.615	15	2	3	0.615	15	2	3	0.615
15	3	1	0.639	15	3	1	0.639	15	3	1	0.639	15	3	1	0.639
15	3	2	0.640	15	3	2	0.608	15	3	2	0.608	15	3	2	0.608
15	3	3	0.641	15	3	3	0.609	15	3	3	0.609	15	3	3	0.609
30	1	1	0.639	30	1	1	0.639	30	1	1	0.639	30	1	1	0.657
30	1	2	0.639	30	1	2	0.617	30	1	2	0.617	30	1	2	0.671
30	1	3	0.639	30	1	3	0.638	30	1	3	0.639	30	1	3	0.671
30	2	1	0.639	30	2	1	0.639	30	2	1	0.639	30	2	1	0.656
30	2	2	0.640	30	2	2	0.640	30	2	2	0.640	30	2	2	0.608
30	2	3	0.640	30	2	3	0.640	30	2	3	0.640	30	2	3	0.608
30	3	1	0.640	30	3	1	0.640	30	3	1	0.640	30	3	1	0.654
30	3	2	0.641	30	3	2	0.641	30	3	2	0.641	30	3	2	0.650
30	3	3	0.641	30	3	3	0.641	30	3	3	0.637	30	3	3	0.643
45	1	1	0.639	45	1	1	0.671	45	1	1	0.671	45	1	1	0.671
45	1	2	0.639	45	1	2	0.671	45	1	2	0.671	45	1	2	0.671
45	1	3	0.639	45	1	3	0.671	45	1	3	0.671	45	1	3	0.671
45	2	1	0.640	45	2	1	0.672	45	2	1	0.672	45	2	1	0.672
45	2	2	0.639	45	2	2	0.662	45	2	2	0.663	45	2	2	0.646
45	2	3	0.640	45	2	3	0.646	45	2	3	0.645	45	2	3	0.645
45	3	1	0.639	45	3	1	0.663	45	3	1	0.664	45	3	1	0.663
45	3	2	0.640	45	3	2	0.649	45	3	2	0.649	45	3	2	0.650
45	3	3	0.641	45	3	3	0.641	45	3	3	0.641	45	3	3	0.641

	Scene 8 $\beta = 90^\circ$					Scene 8 $\beta = 45^\circ$					Scene 8 $\beta = 30^\circ$					Scene 8 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.598		5	1	1	0.598		5	1	1	0.598		5	1	1	0.599	
	5	1	2	0.598		5	1	2	0.596		5	1	2	0.596		5	1	2	0.595	
	5	1	3	0.598		5	1	3	0.597		5	1	3	0.597		5	1	3	0.595	
	5	2	1	0.598		5	2	1	0.598		5	2	1	0.598		5	2	1	0.601	
	5	2	2	0.624		5	2	2	0.622		5	2	2	0.622		5	2	2	0.627	
	5	2	3	0.614		5	2	3	0.611		5	2	3	0.610		5	2	3	0.617	
	5	3	1	0.610		5	3	1	0.610		5	3	1	0.610		5	3	1	0.611	
	5	3	2	0.632		5	3	2	0.628		5	3	2	0.628		5	3	2	0.630	
	5	3	3	0.622		5	3	3	0.619		5	3	3	0.618		5	3	3	0.621	
	15	1	1	0.691		15	1	1	0.691		15	1	1	0.691		15	1	1	0.692	
	15	1	2	0.691		15	1	2	0.686		15	1	2	0.688		15	1	2	0.681	
	15	1	3	0.691		15	1	3	0.687		15	1	3	0.690		15	1	3	0.685	
	15	2	1	0.706		15	2	1	0.711		15	2	1	0.711		15	2	1	0.722	
	15	2	2	0.691		15	2	2	0.693		15	2	2	0.692		15	2	2	0.701	
	15	2	3	0.692		15	2	3	0.692		15	2	3	0.691		15	2	3	0.694	
	15	3	1	0.707		15	3	1	0.706		15	3	1	0.706		15	3	1	0.714	
	15	3	2	0.698		15	3	2	0.700		15	3	2	0.700		15	3	2	0.700	
	15	3	3	0.701		15	3	3	0.701		15	3	3	0.701		15	3	3	0.702	
	30	1	1	0.693		30	1	1	0.693		30	1	1	0.693		30	1	1	0.693	
	30	1	2	0.693		30	1	2	0.692		30	1	2	0.693		30	1	2	0.693	
	30	1	3	0.693		30	1	3	0.691		30	1	3	0.697		30	1	3	0.697	
	30	2	1	0.706		30	2	1	0.722		30	2	1	0.722		30	2	1	0.722	
	30	2	2	0.706		30	2	2	0.714		30	2	2	0.715		30	2	2	0.718	
	30	2	3	0.693		30	2	3	0.694		30	2	3	0.694		30	2	3	0.697	
	30	3	1	0.706		30	3	1	0.712		30	3	1	0.712		30	3	1	0.712	
	30	3	2	0.707		30	3	2	0.714		30	3	2	0.714		30	3	2	0.718	
	30	3	3	0.705		30	3	3	0.708		30	3	3	0.702		30	3	3	0.711	
	45	1	1	0.684		45	1	1	0.684		45	1	1	0.684		45	1	1	0.685	
	45	1	2	0.684		45	1	2	0.712		45	1	2	0.711		45	1	2	0.713	
	45	1	3	0.684		45	1	3	0.718		45	1	3	0.718		45	1	3	0.718	
	45	2	1	0.705		45	2	1	0.705		45	2	1	0.705		45	2	1	0.717	
	45	2	2	0.706		45	2	2	0.706		45	2	2	0.706		45	2	2	0.709	
	45	2	3	0.684		45	2	3	0.684		45	2	3	0.683		45	2	3	0.684	
	45	3	1	0.707		45	3	1	0.712		45	3	1	0.712		45	3	1	0.717	
	45	3	2	0.706		45	3	2	0.714		45	3	2	0.714		45	3	2	0.714	
	45	3	3	0.701		45	3	3	0.703		45	3	3	0.702		45	3	3	0.702	

	Scene 9 $\beta = 90^\circ$					Scene 9 $\beta = 45^\circ$					Scene 9 $\beta = 30^\circ$					Scene 9 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.340		5	1	1	0.341		5	1	1	0.341		5	1	1	0.341	
	5	1	2	0.340		5	1	2	0.329		5	1	2	0.328		5	1	2	0.328	
	5	1	3	0.340		5	1	3	0.324		5	1	3	0.323		5	1	3	0.323	
	5	2	1	0.340		5	2	1	0.340		5	2	1	0.340		5	2	1	0.340	
	5	2	2	0.344		5	2	2	0.343		5	2	2	0.345		5	2	2	0.345	
	5	2	3	0.349		5	2	3	0.342		5	2	3	0.343		5	2	3	0.343	
	5	3	1	0.345		5	3	1	0.344		5	3	1	0.346		5	3	1	0.346	
	5	3	2	0.350		5	3	2	0.359		5	3	2	0.360		5	3	2	0.360	
	5	3	3	0.354		5	3	3	0.359		5	3	3	0.358		5	3	3	0.359	
	15	1	1	0.428		15	1	1	0.428		15	1	1	0.428		15	1	1	0.428	
	15	1	2	0.428		15	1	2	0.407		15	1	2	0.407		15	1	2	0.407	
	15	1	3	0.428		15	1	3	0.407		15	1	3	0.407		15	1	3	0.407	
	15	2	1	0.428		15	2	1	0.428		15	2	1	0.428		15	2	1	0.428	
	15	2	2	0.471		15	2	2	0.447		15	2	2	0.447		15	2	2	0.447	
	15	2	3	0.460		15	2	3	0.437		15	2	3	0.437		15	2	3	0.437	
	15	3	1	0.431		15	3	1	0.431		15	3	1	0.431		15	3	1	0.431	
	15	3	2	0.477		15	3	2	0.453		15	3	2	0.453		15	3	2	0.453	
	15	3	3	0.472		15	3	3	0.448		15	3	3	0.448		15	3	3	0.448	
	30	1	1	0.490		30	1	1	0.490		30	1	1	0.490		30	1	1	0.490	
	30	1	2	0.490		30	1	2	0.466		30	1	2	0.466		30	1	2	0.466	
	30	1	3	0.490		30	1	3	0.466		30	1	3	0.466		30	1	3	0.466	
	30	2	1	0.490		30	2	1	0.491		30	2	1	0.491		30	2	1	0.491	
	30	2	2	0.567		30	2	2	0.595		30	2	2	0.595		30	2	2	0.595	
	30	2	3	0.603		30	2	3	0.633		30	2	3	0.633		30	2	3	0.633	
	30	3	1	0.544		30	3	1	0.549		30	3	1	0.549		30	3	1	0.549	
	30	3	2	0.605		30	3	2	0.635		30	3	2	0.635		30	3	2	0.635	
	30	3	3	0.618		30	3	3	0.649		30	3	3	0.618		30	3	3	0.649	
	45	1	1	0.462		45	1	1	0.463		45	1	1	0.467		45	1	1	0.467	
	45	1	2	0.462		45	1	2	0.452		45	1	2	0.459		45	1	2	0.441	
	45	1	3	0.462		45	1	3	0.485		45	1	3	0.483		45	1	3	0.485	
	45	2	1	0.617		45	2	1	0.617		45	2	1	0.619		45	2	1	0.622	
	45	2	2	0.532		45	2	2	0.541		45	2	2	0.541		45	2	2	0.533	
	45	2	3	0.462		45	2	3	0.463		45	2	3	0.467		45	2	3	0.464	
	45	3	1	0.617		45	3	1	0.617		45	3	1	0.618		45	3	1	0.617	
	45	3	2	0.618		45	3	2	0.619		45	3	2	0.619		45	3	2	0.619	
	45	3	3	0.509		45	3	3	0.503		45	3	3	0.510		45	3	3	0.503	

	Scene 10 $\beta = 90^\circ$					Scene 10 $\beta = 45^\circ$					Scene 10 $\beta = 30^\circ$					Scene 10 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.396		5	1	1	0.396		5	1	1	0.397		5	1	1	0.397	
	5	1	2	0.396		5	1	2	0.389		5	1	2	0.393		5	1	2	0.392	
	5	1	3	0.396		5	1	3	0.389		5	1	3	0.391		5	1	3	0.390	
	5	2	1	0.405		5	2	1	0.405		5	2	1	0.404		5	2	1	0.404	
	5	2	2	0.404		5	2	2	0.392		5	2	2	0.390		5	2	2	0.394	
	5	2	3	0.396		5	2	3	0.380		5	2	3	0.380		5	2	3	0.383	
	5	3	1	0.405		5	3	1	0.405		5	3	1	0.406		5	3	1	0.406	
	5	3	2	0.404		5	3	2	0.390		5	3	2	0.390		5	3	2	0.390	
	5	3	3	0.399		5	3	3	0.383		5	3	3	0.383		5	3	3	0.383	
	15	1	1	0.445		15	1	1	0.445		15	1	1	0.448		15	1	1	0.448	
	15	1	2	0.445		15	1	2	0.423		15	1	2	0.423		15	1	2	0.423	
	15	1	3	0.445		15	1	3	0.423		15	1	3	0.423		15	1	3	0.423	
	15	2	1	0.449		15	2	1	0.449		15	2	1	0.448		15	2	1	0.447	
	15	2	2	0.445		15	2	2	0.423		15	2	2	0.423		15	2	2	0.423	
	15	2	3	0.462		15	2	3	0.439		15	2	3	0.439		15	2	3	0.439	
	15	3	1	0.460		15	3	1	0.460		15	3	1	0.463		15	3	1	0.463	
	15	3	2	0.463		15	3	2	0.440		15	3	2	0.440		15	3	2	0.440	
	15	3	3	0.474		15	3	3	0.450		15	3	3	0.450		15	3	3	0.450	
	30	1	1	0.472		30	1	1	0.472		30	1	1	0.475		30	1	1	0.477	
	30	1	2	0.472		30	1	2	0.448		30	1	2	0.448		30	1	2	0.448	
	30	1	3	0.472		30	1	3	0.448		30	1	3	0.448		30	1	3	0.448	
	30	2	1	0.528		30	2	1	0.528		30	2	1	0.534		30	2	1	0.529	
	30	2	2	0.472		30	2	2	0.472		30	2	2	0.492		30	2	2	0.496	
	30	2	3	0.502		30	2	3	0.502		30	2	3	0.509		30	2	3	0.527	
	30	3	1	0.538		30	3	1	0.538		30	3	1	0.540		30	3	1	0.536	
	30	3	2	0.523		30	3	2	0.528		30	3	2	0.549		30	3	2	0.549	
	30	3	3	0.529		30	3	3	0.531		30	3	3	0.534		30	3	3	0.555	
	45	1	1	0.540		45	1	1	0.540		45	1	1	0.540		45	1	1	0.540	
	45	1	2	0.540		45	1	2	0.540		45	1	2	0.540		45	1	2	0.520	
	45	1	3	0.540		45	1	3	0.544		45	1	3	0.543		45	1	3	0.567	
	45	2	1	0.545		45	2	1	0.549		45	2	1	0.549		45	2	1	0.549	
	45	2	2	0.544		45	2	2	0.548		45	2	2	0.548		45	2	2	0.522	
	45	2	3	0.540		45	2	3	0.567		45	2	3	0.567		45	2	3	0.566	
	45	3	1	0.546		45	3	1	0.551		45	3	1	0.551		45	3	1	0.551	
	45	3	2	0.546		45	3	2	0.549		45	3	2	0.549		45	3	2	0.549	
	45	3	3	0.547		45	3	3	0.554		45	3	3	0.554		45	3	3	0.554	

Scene 11 $\beta = 90^\circ$				Scene 11 $\beta = 45^\circ$				Scene 11 $\beta = 30^\circ$				Scene 11 $\beta = 15^\circ$			
α	r	s	IoU	α	r	s	IoU	α	r	s	IoU	α	r	s	IoU
5	1	1	0.531	5	1	1	0.536	5	1	1	0.536	5	1	1	0.536
5	1	2	0.531	5	1	2	0.558	5	1	2	0.558	5	1	2	0.558
5	1	3	0.531	5	1	3	0.558	5	1	3	0.558	5	1	3	0.558
5	2	1	0.531	5	2	1	0.535	5	2	1	0.535	5	2	1	0.535
5	2	2	0.542	5	2	2	0.569	5	2	2	0.569	5	2	2	0.569
5	2	3	0.547	5	2	3	0.574	5	2	3	0.574	5	2	3	0.574
5	3	1	0.542	5	3	1	0.542	5	3	1	0.542	5	3	1	0.542
5	3	2	0.552	5	3	2	0.580	5	3	2	0.580	5	3	2	0.580
5	3	3	0.556	5	3	3	0.584	5	3	3	0.584	5	3	3	0.584
15	1	1	0.636	15	1	1	0.647	15	1	1	0.647	15	1	1	0.647
15	1	2	0.636	15	1	2	0.668	15	1	2	0.668	15	1	2	0.668
15	1	3	0.636	15	1	3	0.668	15	1	3	0.668	15	1	3	0.668
15	2	1	0.636	15	2	1	0.638	15	2	1	0.638	15	2	1	0.638
15	2	2	0.680	15	2	2	0.714	15	2	2	0.714	15	2	2	0.714
15	2	3	0.662	15	2	3	0.695	15	2	3	0.695	15	2	3	0.695
15	3	1	0.658	15	3	1	0.656	15	3	1	0.658	15	3	1	0.658
15	3	2	0.686	15	3	2	0.720	15	3	2	0.720	15	3	2	0.720
15	3	3	0.669	15	3	3	0.702	15	3	3	0.702	15	3	3	0.702
30	1	1	0.670	30	1	1	0.688	30	1	1	0.688	30	1	1	0.688
30	1	2	0.670	30	1	2	0.704	30	1	2	0.704	30	1	2	0.704
30	1	3	0.670	30	1	3	0.693	30	1	3	0.693	30	1	3	0.693
30	2	1	0.686	30	2	1	0.686	30	2	1	0.690	30	2	1	0.695
30	2	2	0.687	30	2	2	0.690	30	2	2	0.721	30	2	2	0.653
30	2	3	0.670	30	2	3	0.669	30	2	3	0.693	30	2	3	0.637
30	3	1	0.687	30	3	1	0.687	30	3	1	0.689	30	3	1	0.689
30	3	2	0.687	30	3	2	0.687	30	3	2	0.721	30	3	2	0.653
30	3	3	0.677	30	3	3	0.677	30	3	3	0.682	30	3	3	0.643
45	1	1	0.669	45	1	1	0.697	45	1	1	0.695	45	1	1	0.697
45	1	2	0.669	45	1	2	0.702	45	1	2	0.702	45	1	2	0.636
45	1	3	0.669	45	1	3	0.695	45	1	3	0.695	45	1	3	0.636
45	2	1	0.687	45	2	1	0.687	45	2	1	0.687	45	2	1	0.681
45	2	2	0.669	45	2	2	0.669	45	2	2	0.672	45	2	2	0.636
45	2	3	0.669	45	2	3	0.670	45	2	3	0.690	45	2	3	0.673
45	3	1	0.687	45	3	1	0.687	45	3	1	0.690	45	3	1	0.687
45	3	2	0.682	45	3	2	0.687	45	3	2	0.688	45	3	2	0.686
45	3	3	0.681	45	3	3	0.687	45	3	3	0.684	45	3	3	0.686

	Scene 12 $\beta = 90^\circ$					Scene 12 $\beta = 45^\circ$					Scene 12 $\beta = 30^\circ$					Scene 12 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.576		5	1	1	0.577		5	1	1	0.577		5	1	1	0.577	
	5	1	2	0.576		5	1	2	0.605		5	1	2	0.605		5	1	2	0.605	
	5	1	3	0.576		5	1	3	0.605		5	1	3	0.605		5	1	3	0.605	
	5	2	1	0.578		5	2	1	0.578		5	2	1	0.577		5	2	1	0.577	
	5	2	2	0.576		5	2	2	0.597		5	2	2	0.596		5	2	2	0.596	
	5	2	3	0.577		5	2	3	0.601		5	2	3	0.600		5	2	3	0.600	
	5	3	1	0.576		5	3	1	0.576		5	3	1	0.576		5	3	1	0.576	
	5	3	2	0.577		5	3	2	0.591		5	3	2	0.590		5	3	2	0.590	
	5	3	3	0.577		5	3	3	0.590		5	3	3	0.592		5	3	3	0.592	
	15	1	1	0.576		15	1	1	0.576		15	1	1	0.578		15	1	1	0.578	
	15	1	2	0.576		15	1	2	0.572		15	1	2	0.576		15	1	2	0.576	
	15	1	3	0.576		15	1	3	0.604		15	1	3	0.602		15	1	3	0.602	
	15	2	1	0.594		15	2	1	0.594		15	2	1	0.591		15	2	1	0.591	
	15	2	2	0.584		15	2	2	0.555		15	2	2	0.555		15	2	2	0.555	
	15	2	3	0.576		15	2	3	0.547		15	2	3	0.547		15	2	3	0.547	
	15	3	1	0.594		15	3	1	0.594		15	3	1	0.592		15	3	1	0.592	
	15	3	2	0.589		15	3	2	0.560		15	3	2	0.560		15	3	2	0.560	
	15	3	3	0.576		15	3	3	0.547		15	3	3	0.547		15	3	3	0.547	
	30	1	1	0.587		30	1	1	0.587		30	1	1	0.592		30	1	1	0.592	
	30	1	2	0.587		30	1	2	0.558		30	1	2	0.558		30	1	2	0.558	
	30	1	3	0.587		30	1	3	0.587		30	1	3	0.580		30	1	3	0.580	
	30	2	1	0.593		30	2	1	0.593		30	2	1	0.588		30	2	1	0.588	
	30	2	2	0.593		30	2	2	0.593		30	2	2	0.623		30	2	2	0.623	
	30	2	3	0.594		30	2	3	0.594		30	2	3	0.624		30	2	3	0.624	
	30	3	1	0.594		30	3	1	0.596		30	3	1	0.590		30	3	1	0.591	
	30	3	2	0.587		30	3	2	0.583		30	3	2	0.616		30	3	2	0.616	
	30	3	3	0.593		30	3	3	0.599		30	3	3	0.587		30	3	3	0.623	
	45	1	1	0.556		45	1	1	0.556		45	1	1	0.559		45	1	1	0.559	
	45	1	2	0.556		45	1	2	0.574		45	1	2	0.584		45	1	2	0.528	
	45	1	3	0.556		45	1	3	0.568		45	1	3	0.563		45	1	3	0.528	
	45	2	1	0.556		45	2	1	0.556		45	2	1	0.554		45	2	1	0.554	
	45	2	2	0.570		45	2	2	0.570		45	2	2	0.571		45	2	2	0.542	
	45	2	3	0.572		45	2	3	0.573		45	2	3	0.573		45	2	3	0.573	
	45	3	1	0.561		45	3	1	0.563		45	3	1	0.563		45	3	1	0.563	
	45	3	2	0.568		45	3	2	0.574		45	3	2	0.573		45	3	2	0.573	
	45	3	3	0.575		45	3	3	0.575		45	3	3	0.576		45	3	3	0.576	

	Scene 13 $\beta = 90^\circ$					Scene 13 $\beta = 45^\circ$					Scene 13 $\beta = 30^\circ$					Scene 13 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.614		5	1	1	0.614		5	1	1	0.614		5	1	1	0.614	
	5	1	2	0.614		5	1	2	0.645		5	1	2	0.645		5	1	2	0.645	
	5	1	3	0.614		5	1	3	0.645		5	1	3	0.645		5	1	3	0.645	
	5	2	1	0.614		5	2	1	0.614		5	2	1	0.614		5	2	1	0.614	
	5	2	2	0.635		5	2	2	0.667		5	2	2	0.667		5	2	2	0.667	
	5	2	3	0.635		5	2	3	0.667		5	2	3	0.667		5	2	3	0.667	
	5	3	1	0.620		5	3	1	0.620		5	3	1	0.620		5	3	1	0.620	
	5	3	2	0.647		5	3	2	0.679		5	3	2	0.679		5	3	2	0.679	
	5	3	3	0.642		5	3	3	0.674		5	3	3	0.674		5	3	3	0.674	
	15	1	1	0.682		15	1	1	0.685		15	1	1	0.686		15	1	1	0.686	
	15	1	2	0.682		15	1	2	0.716		15	1	2	0.716		15	1	2	0.716	
	15	1	3	0.682		15	1	3	0.716		15	1	3	0.716		15	1	3	0.716	
	15	2	1	0.704		15	2	1	0.708		15	2	1	0.708		15	2	1	0.708	
	15	2	2	0.706		15	2	2	0.741		15	2	2	0.741		15	2	2	0.741	
	15	2	3	0.682		15	2	3	0.716		15	2	3	0.716		15	2	3	0.716	
	15	3	1	0.706		15	3	1	0.706		15	3	1	0.706		15	3	1	0.706	
	15	3	2	0.707		15	3	2	0.742		15	3	2	0.742		15	3	2	0.742	
	15	3	3	0.691		15	3	3	0.726		15	3	3	0.726		15	3	3	0.726	
	30	1	1	0.697		30	1	1	0.701		30	1	1	0.704		30	1	1	0.704	
	30	1	2	0.697		30	1	2	0.732		30	1	2	0.732		30	1	2	0.732	
	30	1	3	0.697		30	1	3	0.662		30	1	3	0.710		30	1	3	0.717	
	30	2	1	0.704		30	2	1	0.704		30	2	1	0.712		30	2	1	0.712	
	30	2	2	0.706		30	2	2	0.736		30	2	2	0.697		30	2	2	0.671	
	30	2	3	0.697		30	2	3	0.732		30	2	3	0.732		30	2	3	0.662	
	30	3	1	0.706		30	3	1	0.706		30	3	1	0.705		30	3	1	0.706	
	30	3	2	0.707		30	3	2	0.721		30	3	2	0.716		30	3	2	0.672	
	30	3	3	0.703		30	3	3	0.716		30	3	3	0.702		30	3	3	0.668	
	45	1	1	0.703		45	1	1	0.738		45	1	1	0.738		45	1	1	0.738	
	45	1	2	0.703		45	1	2	0.710		45	1	2	0.726		45	1	2	0.668	
	45	1	3	0.703		45	1	3	0.738		45	1	3	0.738		45	1	3	0.668	
	45	2	1	0.703		45	2	1	0.735		45	2	1	0.738		45	2	1	0.738	
	45	2	2	0.706		45	2	2	0.708		45	2	2	0.711		45	2	2	0.671	
	45	2	3	0.706		45	2	3	0.710		45	2	3	0.712		45	2	3	0.710	
	45	3	1	0.707		45	3	1	0.729		45	3	1	0.729		45	3	1	0.729	
	45	3	2	0.703		45	3	2	0.709		45	3	2	0.712		45	3	2	0.714	
	45	3	3	0.707		45	3	3	0.713		45	3	3	0.718		45	3	3	0.718	

	Scene 14 $\beta = 90^\circ$					Scene 14 $\beta = 45^\circ$					Scene 14 $\beta = 30^\circ$					Scene 14 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.519		5	1	1	0.519		5	1	1	0.519		5	1	1	0.519	
	5	1	2	0.519		5	1	2	0.521		5	1	2	0.521		5	1	2	0.516	
	5	1	3	0.519		5	1	3	0.545		5	1	3	0.545		5	1	3	0.545	
	5	2	1	0.519		5	2	1	0.519		5	2	1	0.519		5	2	1	0.519	
	5	2	2	0.528		5	2	2	0.554		5	2	2	0.554		5	2	2	0.554	
	5	2	3	0.532		5	2	3	0.559		5	2	3	0.559		5	2	3	0.559	
	5	3	1	0.526		5	3	1	0.526		5	3	1	0.526		5	3	1	0.526	
	5	3	2	0.534		5	3	2	0.561		5	3	2	0.561		5	3	2	0.561	
	5	3	3	0.538		5	3	3	0.565		5	3	3	0.565		5	3	3	0.565	
	15	1	1	0.504		15	1	1	0.504		15	1	1	0.505		15	1	1	0.505	
	15	1	2	0.504		15	1	2	0.529		15	1	2	0.529		15	1	2	0.529	
	15	1	3	0.504		15	1	3	0.529		15	1	3	0.529		15	1	3	0.529	
	15	2	1	0.546		15	2	1	0.546		15	2	1	0.546		15	2	1	0.544	
	15	2	2	0.533		15	2	2	0.560		15	2	2	0.560		15	2	2	0.560	
	15	2	3	0.504		15	2	3	0.529		15	2	3	0.529		15	2	3	0.529	
	15	3	1	0.542		15	3	1	0.542		15	3	1	0.543		15	3	1	0.542	
	15	3	2	0.544		15	3	2	0.571		15	3	2	0.571		15	3	2	0.571	
	15	3	3	0.521		15	3	3	0.547		15	3	3	0.547		15	3	3	0.547	
	30	1	1	0.496		30	1	1	0.496		30	1	1	0.496		30	1	1	0.496	
	30	1	2	0.496		30	1	2	0.521		30	1	2	0.521		30	1	2	0.521	
	30	1	3	0.496		30	1	3	0.503		30	1	3	0.504		30	1	3	0.491	
	30	2	1	0.545		30	2	1	0.545		30	2	1	0.545		30	2	1	0.547	
	30	2	2	0.535		30	2	2	0.535		30	2	2	0.533		30	2	2	0.508	
	30	2	3	0.496		30	2	3	0.496		30	2	3	0.492		30	2	3	0.471	
	30	3	1	0.546		30	3	1	0.546		30	3	1	0.546		30	3	1	0.546	
	30	3	2	0.546		30	3	2	0.546		30	3	2	0.546		30	3	2	0.519	
	30	3	3	0.522		30	3	3	0.522		30	3	3	0.525		30	3	3	0.496	
	45	1	1	0.484		45	1	1	0.508		45	1	1	0.508		45	1	1	0.508	
	45	1	2	0.484		45	1	2	0.508		45	1	2	0.508		45	1	2	0.460	
	45	1	3	0.484		45	1	3	0.508		45	1	3	0.508		45	1	3	0.460	
	45	2	1	0.543		45	2	1	0.548		45	2	1	0.547		45	2	1	0.548	
	45	2	2	0.545		45	2	2	0.535		45	2	2	0.544		45	2	2	0.518	
	45	2	3	0.484		45	2	3	0.465		45	2	3	0.484		45	2	3	0.468	
	45	3	1	0.545		45	3	1	0.545		45	3	1	0.544		45	3	1	0.545	
	45	3	2	0.539		45	3	2	0.538		45	3	2	0.534		45	3	2	0.538	
	45	3	3	0.513		45	3	3	0.518		45	3	3	0.517		45	3	3	0.517	

	Scene 15 $\beta = 90^\circ$					Scene 15 $\beta = 45^\circ$					Scene 15 $\beta = 30^\circ$					Scene 15 $\beta = 15^\circ$				
	α	r	s	IoU		α	r	s	IoU		α	r	s	IoU		α	r	s	IoU	
	5	1	1	0.500		5	1	1	0.504		5	1	1	0.504		5	1	1	0.504	
	5	1	2	0.500		5	1	2	0.512		5	1	2	0.509		5	1	2	0.512	
	5	1	3	0.500		5	1	3	0.522		5	1	3	0.520		5	1	3	0.522	
	5	2	1	0.500		5	2	1	0.500		5	2	1	0.502		5	2	1	0.500	
	5	2	2	0.510		5	2	2	0.519		5	2	2	0.526		5	2	2	0.519	
	5	2	3	0.513		5	2	3	0.522		5	2	3	0.530		5	2	3	0.521	
	5	3	1	0.507		5	3	1	0.507		5	3	1	0.507		5	3	1	0.507	
	5	3	2	0.516		5	3	2	0.526		5	3	2	0.528		5	3	2	0.526	
	5	3	3	0.515		5	3	3	0.525		5	3	3	0.527		5	3	3	0.524	
	15	1	1	0.489		15	1	1	0.503		15	1	1	0.501		15	1	1	0.503	
	15	1	2	0.489		15	1	2	0.507		15	1	2	0.500		15	1	2	0.509	
	15	1	3	0.489		15	1	3	0.502		15	1	3	0.496		15	1	3	0.504	
	15	2	1	0.521		15	2	1	0.521		15	2	1	0.525		15	2	1	0.521	
	15	2	2	0.518		15	2	2	0.524		15	2	2	0.523		15	2	2	0.524	
	15	2	3	0.489		15	2	3	0.495		15	2	3	0.495		15	2	3	0.495	
	15	3	1	0.521		15	3	1	0.521		15	3	1	0.523		15	3	1	0.521	
	15	3	2	0.521		15	3	2	0.524		15	3	2	0.516		15	3	2	0.525	
	15	3	3	0.501		15	3	3	0.505		15	3	3	0.502		15	3	3	0.505	
	30	1	1	0.481		30	1	1	0.505		30	1	1	0.502		30	1	1	0.505	
	30	1	2	0.481		30	1	2	0.502		30	1	2	0.501		30	1	2	0.503	
	30	1	3	0.481		30	1	3	0.485		30	1	3	0.483		30	1	3	0.487	
	30	2	1	0.521		30	2	1	0.525		30	2	1	0.529		30	2	1	0.525	
	30	2	2	0.517		30	2	2	0.543		30	2	2	0.519		30	2	2	0.541	
	30	2	3	0.481		30	2	3	0.505		30	2	3	0.498		30	2	3	0.505	
	30	3	1	0.521		30	3	1	0.521		30	3	1	0.523		30	3	1	0.523	
	30	3	2	0.521		30	3	2	0.519		30	3	2	0.517		30	3	2	0.519	
	30	3	3	0.502		30	3	3	0.503		30	3	3	0.504		30	3	3	0.502	
	45	1	1	0.472		45	1	1	0.496		45	1	1	0.496		45	1	1	0.496	
	45	1	2	0.472		45	1	2	0.496		45	1	2	0.496		45	1	2	0.496	
	45	1	3	0.472		45	1	3	0.496		45	1	3	0.496		45	1	3	0.496	
	45	2	1	0.520		45	2	1	0.520		45	2	1	0.523		45	2	1	0.520	
	45	2	2	0.520		45	2	2	0.520		45	2	2	0.519		45	2	2	0.516	
	45	2	3	0.472		45	2	3	0.472		45	2	3	0.472		45	2	3	0.472	
	45	3	1	0.521		45	3	1	0.526		45	3	1	0.525		45	3	1	0.526	
	45	3	2	0.520		45	3	2	0.520		45	3	2	0.514		45	3	2	0.521	
	45	3	3	0.494		45	3	3	0.494		45	3	3	0.498		45	3	3	0.497	

Scene 16 $\beta = 90^\circ$				Scene 16 $\beta = 45^\circ$				Scene 16 $\beta = 30^\circ$				Scene 16 $\beta = 15^\circ$			
α	r	s	IoU	α	r	s	IoU	α	r	s	IoU	α	r	s	IoU
5	1	1	0.690	5	1	1	0.690	5	1	1	0.690	5	1	1	0.690
5	1	2	0.690	5	1	2	0.694	5	1	2	0.694	5	1	2	0.694
5	1	3	0.690	5	1	3	0.698	5	1	3	0.698	5	1	3	0.698
5	2	1	0.690	5	2	1	0.691	5	2	1	0.691	5	2	1	0.691
5	2	2	0.698	5	2	2	0.686	5	2	2	0.686	5	2	2	0.686
5	2	3	0.698	5	2	3	0.690	5	2	3	0.690	5	2	3	0.689
5	3	1	0.693	5	3	1	0.693	5	3	1	0.693	5	3	1	0.693
5	3	2	0.699	5	3	2	0.686	5	3	2	0.686	5	3	2	0.687
5	3	3	0.700	5	3	3	0.691	5	3	3	0.691	5	3	3	0.690
15	1	1	0.722	15	1	1	0.722	15	1	1	0.722	15	1	1	0.722
15	1	2	0.722	15	1	2	0.717	15	1	2	0.718	15	1	2	0.718
15	1	3	0.722	15	1	3	0.730	15	1	3	0.730	15	1	3	0.730
15	2	1	0.723	15	2	1	0.722	15	2	1	0.722	15	2	1	0.722
15	2	2	0.722	15	2	2	0.688	15	2	2	0.688	15	2	2	0.686
15	2	3	0.723	15	2	3	0.700	15	2	3	0.695	15	2	3	0.690
15	3	1	0.723	15	3	1	0.723	15	3	1	0.723	15	3	1	0.723
15	3	2	0.724	15	3	2	0.700	15	3	2	0.701	15	3	2	0.700
15	3	3	0.724	15	3	3	0.705	15	3	3	0.703	15	3	3	0.700
30	1	1	0.722	30	1	1	0.722	30	1	1	0.722	30	1	1	0.722
30	1	2	0.722	30	1	2	0.710	30	1	2	0.711	30	1	2	0.711
30	1	3	0.722	30	1	3	0.753	30	1	3	0.747	30	1	3	0.747
30	2	1	0.722	30	2	1	0.722	30	2	1	0.722	30	2	1	0.725
30	2	2	0.723	30	2	2	0.723	30	2	2	0.723	30	2	2	0.719
30	2	3	0.723	30	2	3	0.723	30	2	3	0.720	30	2	3	0.711
30	3	1	0.723	30	3	1	0.727	30	3	1	0.727	30	3	1	0.727
30	3	2	0.723	30	3	2	0.699	30	3	2	0.699	30	3	2	0.718
30	3	3	0.724	30	3	3	0.707	30	3	3	0.724	30	3	3	0.716
45	1	1	0.721	45	1	1	0.732	45	1	1	0.729	45	1	1	0.732
45	1	2	0.721	45	1	2	0.757	45	1	2	0.757	45	1	2	0.708
45	1	3	0.721	45	1	3	0.730	45	1	3	0.730	45	1	3	0.686
45	2	1	0.721	45	2	1	0.716	45	2	1	0.720	45	2	1	0.721
45	2	2	0.723	45	2	2	0.746	45	2	2	0.730	45	2	2	0.708
45	2	3	0.722	45	2	3	0.748	45	2	3	0.730	45	2	3	0.748
45	3	1	0.722	45	3	1	0.711	45	3	1	0.721	45	3	1	0.711
45	3	2	0.723	45	3	2	0.727	45	3	2	0.723	45	3	2	0.728
45	3	3	0.724	45	3	3	0.737	45	3	3	0.724	45	3	3	0.742